

Aodong Liu

Computational Chemistry | Nuclear Quantum Effects | Scientific HPC

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Research Profile

Computational chemist developing nuclear–electronic orbital theory, scalable electronic-structure algorithms, and coupled nuclear/electronic quantum dynamics. Lead NEO-module developer for Gaussian 25 and active scientific-software developer in Fortran and C++, with eight peer-reviewed journal articles (first-listed author on three and co-first author on a fourth) and extensive MPI/OpenMP/HPC experience.

Appointments

University of Washington

Postdoctoral Scholar, Li Group

Seattle, WA

Aug. 2026–Present

- Continue development of scalable quantum-chemistry and coupled nuclear/electronic dynamics methods.

University of Washington

Graduate Research Assistant, Li Group

Seattle, WA

Sep. 2021–Jun. 2026

- Developed NEO electronic-structure and non-Born–Oppenheimer dynamics methods and their scalable scientific-software implementations.
- Co-developed a distributed Cholesky framework demonstrated on 52 AMD EPYC nodes/104 MPI ranks (9,984 bound CPU cores), enabling a 500-quantum-proton calculation and 96.2-second full NEO Fock build.

Gaussian, Inc.

Software Developer

Wallingford, CT

Feb. 2023; Apr. and Sep. 2024

- Led development of the Nuclear–Electronic Orbital module in Gaussian 25.

University of Washington

Teaching Assistant, Graduate Computational Chemistry (CHEM 465)

Seattle, WA

Winter 2022, 2023

Education

University of Washington

Ph.D., Theoretical Chemistry; Advisor: Xiaosong Li

Certificate in Quantum Information Science and Engineering.

Seattle, WA

Jun. 2026

Vanderbilt University

B.A., Physical Chemistry with Honors; Minor, Scientific Computing

Undergraduate advisor: Jens Meiler. Thesis: *de novo* modeling of CNIH1 and its docking mechanism with the AMPA receptor.

Nashville, TN

May 2020

Technical Expertise

Programming: C++, Fortran, Python, Bash/Shell; C, CUDA, Java, MATLAB.

HPC/software: MPI, OpenMP, distributed-memory algorithms, CMake, Git, Linux, CI/CD, Gaussian, ChronusQ.

Methods: NEO theory, electronic structure, DFT, Cholesky decomposition, relativistic methods, real-time and quasiclassical dynamics.

Peer-Reviewed Publications

1. Tang, D.; **Liu, A.**; Culpitt, T.; Li, X. “Ab Initio Two-Spinor Ehrenfest Dynamics with Magnetic Field.” *Phys. Chem. Chem. Phys.* (2026), published online. doi:10.1039/D6CP02263D.
2. Yang, M.; Upadhyay, S.; **Liu, A.**; Li, X. “Spin–Orbit-Resolved Strong-Field Ionization from Real-Time Relativistic Dynamics.” *J. Chem. Phys.* **165**, 014105 (2026). doi:10.1063/5.0320154.
3. **Liu, A.**; Zhang, T.; Lambros, E.; Hammes-Schiffer, S.; Li, X. “Nuclear–Electronic Orbital Method at Scale: A Two-Stage Hybrid-Layout Distributed Cholesky Decomposition Approach.” *APL Computational Physics* **2**, 026109 (2026). Editor’s Pick. doi:10.1063/5.0318199.
4. Aieta, C.; Garner, S. M.; **Liu, A.**; Li, X.; Hammes-Schiffer, S. “Nuclear–Electronic Orbital Quasiclassical Trajectory Method for Vibrational Spectroscopy.” *J. Chem. Phys.* **164**, 144123 (2026). doi:10.1063/5.0317500.

Peer-Reviewed Publications (continued)

5. Tang, D.*; **Liu, A.***; Culpitt, T.; Hammes-Schiffer, S.; Li, X. “Simulating Magnetic Field-Driven Real-Time Quantum Dynamics Using London Nuclear–Electronic Orbital Approach.” *J. Chem. Theory Comput.* **21**, 4357–4364 (2025). doi:10.1021/acs.jctc.5c00273.
6. Kovtun, M.; Lambros, E.; **Liu, A.**; Tang, D.; Williams-Young, D. B.; Li, X. “Accelerating Relativistic Exact-Two-Component Density Functional Theory Calculations with Graphical Processing Units.” *J. Chem. Theory Comput.* **20**, 7694–7699 (2024). doi:10.1021/acs.jctc.4c00843.
7. **Liu, A.**; Zhang, T.; Hammes-Schiffer, S.; Li, X. “Multicomponent Cholesky Decomposition: Application to Nuclear–Electronic Orbital Theory.” *J. Chem. Theory Comput.* **19**, 6255–6262 (2023). doi:10.1021/acs.jctc.3c00686.
8. **Liu, A.**; Chow, M.; Wildman, A.; Frisch, M. J.; Hammes-Schiffer, S.; Li, X. “Simultaneous Optimization of Nuclear–Electronic Orbitals.” *J. Phys. Chem. A* **126**, 7033–7039 (2022). doi:10.1021/acs.jpca.2c05172.

*Equal contribution.

Pending Manuscripts

1. Zhang, T.; **Liu, A.**; Upadhyay, S.; Hu, H.; Beck, R. A.; Bersson, J.; Culpitt, T.; Garner, S.; Kovtun, M.; Li, R. R.; Liao, C.; Lu, L.; Majumder, R.; Oele, M.; Peltekis, A. A.; Petrone, A.; Shayit, A.; Sun, S.; Tang, D.; Wang, T.; Wildman, A.; Williams-Young, D.; Yuwono, S. H.; Zhao, L.; Lambros, E.; Hoyer, C.; Valeev, E. F.; DePrince, A. E., III; Hammes-Schiffer, S.; Li, X. “Chronus Quantum 1.0: A Unified Framework for Relativistic, Time-Dependent, and Multicomponent Electronic Structure Theory.” Submitted, 2026.
2. **Liu, A.**; Beck, R.; Upadhyay, S.; Garner, S. M.; Graham, T.; Ghandi, K.; Pearce, C.; Hammes-Schiffer, S.; Li, X. “Vibrational Bonding in Br μ -Br from Correlated Nuclear–Electronic Wavefunctions.” In preparation, 2026.
3. **Liu, A.**; Lambros, E.; Beck, R.; Hammes-Schiffer, S.; Li, X. “Forming the Bound Electron: Non-Equilibrium Electronic Ionization Dynamics of Water Clusters.” In preparation, 2026.
4. Zheng, J.; **Liu, A.**; Frisch, M. J. “A Tutorial for Nuclear-Electronic Orbital (NEO) Method in the Gaussian Program.” In preparation, 2026.

Selected Presentations

- ACS Spring 2026, oral: “Large-Scale Real-Time NEO–TDDFT Simulations of Ultrafast Solvated Electron Dynamics,” Atlanta, March 2026.
- IDREAM EFRC All Hands Meeting, PNNL, October 2025.
- IDREAM EFRC Meeting, oral: “Ultrafast Processes After Low-Energy Electron Attachment in Water Clusters,” PNNL, October 2024.
- ACS Spring 2024, oral: “Multicomponent Cholesky Decomposition: Application to Nuclear–Electronic Orbital Theory,” New Orleans, March 2024.
- EFRC–Hub–CMS–CCS PI Meeting, poster: “Multiscale Nuclear–Electronic Orbital Quantum Dynamics in Complex Environments,” virtual, September 2023.
- EFRC–Hub–CMS–CCS PI Meeting, poster: “Radiolysis in Extreme Environments,” virtual, September 2023.
- IDREAM EFRC Meeting, oral: “A Computational Study of Potassium K-Edge XANES Spectra,” PNNL, August 2023.
- SciDAC Summit, oral: “Simultaneous Optimization of Nuclear–Electronic Orbitals,” LBNL, December 2022.
- Vanderbilt Student Research Fair, oral: “Structural Modeling and Validation of CNIH1 Protein,” Nashville, September 2019.

Awards

- Excellence in Chemistry Graduate Fellowship Award, University of Washington, 2021.
- Thomas W. Martin Award, Vanderbilt University, 2020; Dr. Sidney & Becca Fleischer Award, 2019.
- Vanderbilt Undergraduate Summer Research Program Fellowship, 2018.

Leadership and Service

- Hyak Governance Board member and Research Computing Club officer, University of Washington, since 2023; RCC mentor since 2022.
- Engineering Discovery Days volunteer: developed and delivered hands-on K–12 demonstrations of superconductivity, quantum levitation, and MAGLEV principles; guided visitors through the underlying physical concepts.